

Multiple Choice

1. **Answer:** E

Options: A) 64 ft/sec; B) 160 ft/sec; C) 80 ft/sec; D) 256 ft/sec; E) 128 ft/sec

Note: Correct option: E - 128 ft/sec

2. **Answer:** A

Options: A) converted into matter; B) changed to a different form; C) transferred to the surrounding environment; D) created within the water; E) both absorbed and released simultaneously

Note: Correct option: A - converted into matter

3. **Answer:** A

Options: A) 1.60; B) 1.22; C) 0.98; D) 1.45; E) 1.05

Note: Correct option: A - 1.60

4. **Answer:** A

Options: A) - 0.50 m/sec; B) - 0.52 m/sec; C) - 0.45 m/sec; D) - 0.38 m/sec; E) - 0.60 m/sec

Note: Correct option: A - - 0.50 m/sec

5. **Answer:** C

Options: A) Both horizontal and vertical; B) Diagonal; C) horizontal; D) Angular; E) Radial

Note: Correct option: C - horizontal

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '390 K'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Neon'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 288 newtons. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 300 N. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key